



Operation Guide

SIGMATRONIC IR Industrial Battery Charger Quick Start Procedure





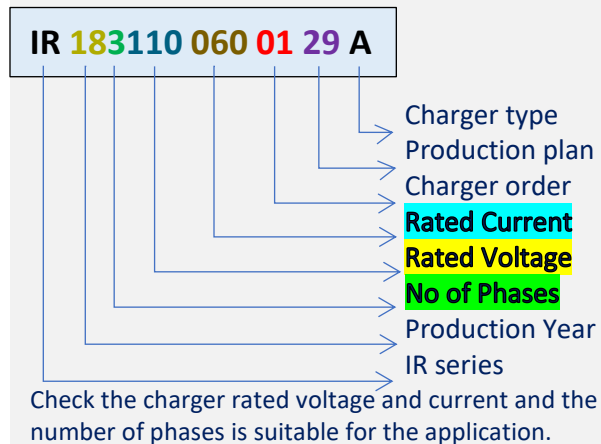
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OVERVIEW

The **IR series** provides the fully protection for DC critical industrial systems from unexpected side effects of power obstructions. It relies on microcontroller control to guarantee the needed flexibility and stability for different applications and loads. The following procedure is the wiring and operating of the charger.

Step 1: Model Identification

To verify the Charger unit is the correct one for the application you need, check its type, output ratings on the nameplate charger. This can be also identified from the serial number partitions as shown in this example.



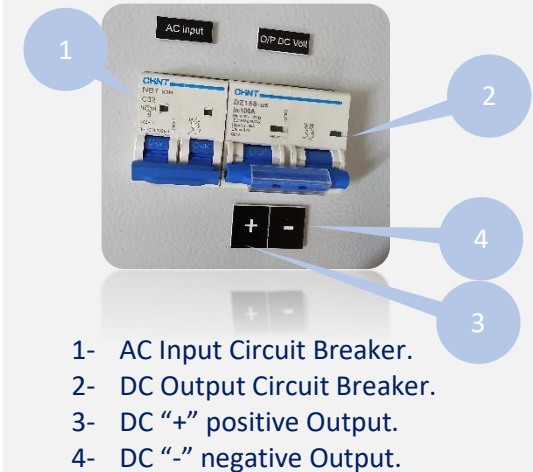
Step 2: Prepare for Installation

The charger must be mounted in a vertical position only.
Avoid mounting close to high heat sources.
The mounting location must be free from frost.
Do not restrict the flow of air through the air filters or inlets and outlets of air within the charger.
Correct air clearance around the charger must be observed.
Cable from the back.
AC and DC cable connected to the charger should be sized properly according to the amperage.

Step 3: Installation

Follow the standard installation procedure carefully before start up the battery charger. Ignoring the strict following to the instructions may lead to serious situations such personnel injury, property damage or equipment failure.
Make sure that main switch (within the charger or external) are in "off position".
Connect the ac source to the input circuit breaker.
Turn on the input circuit breaker.
Check the display and the displayed voltage on it.
Switch off the charger then plugin the dc cables, check the polarization.
Switch on the charger and check the current on the display.
Use suitable tools to tie all the cables inside the terminals.
-WARNING! HIGH EARTH LEAKAGE CURRENT: Earth connection is essential before connecting power inputs & outputs!

Step 4: Connection Labeling



Step 3: Installation



Sales & Head Office and Factory:

📍 Egypt, Cairo, El Obour City, Area No 22, 23, Block 13037, 1st Industrial zone.

www.sigma-engineer.com

✉ Info@sigma-engineer.com



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Step 4: Applying Power and Front panel



Mode	Function*
Normal	-The upper display shows the voltage value. -The lower display shows the current value. -The "ON-OFF" LED is ON. -The "Charge" LED is ON. -The "Limit" LED is OFF.
Current Limit1	-The upper display shows the voltage value. -The lower display swing between current value and "CrL1" -The "ON-OFF" LED is ON. -The "Charge" LED is ON. -The "Limit" LED is ON.
Current Limit2	-The upper display shows the voltage value. -The lower display swing between current value and "CrL2" -The "ON-OFF" LED is ON. -The "Charge" LED is ON. -The "Limit" LED is ON.
Overt Voltage	-The upper display shows "E-OV". -The lower display displays "----" -The "ON-OFF" LED is OFF. -The "Charge" LED is OFF. -The "Limit" LED is OFF.
Overt Load	-The upper display shows "----". -The lower display shows "E-OL" -The "ON-OFF" LED is OFF. -The "Charge" LED is OFF. -The "Limit" LED is OFF.
Short Circuit	-The upper display shows "----". -The lower display shows "E-OL" -The "ON-OFF" LED is OFF. -The "Charge" LED is OFF. -The "Limit" LED is OFF.
Setup Mode	-To enter Setup Mode, press the Enter Button for 2 second. - in setup mode you can change the parameters from 0~4 • "Pr00": AUTO Voltage • "Pr01": Manual Voltage. • "Pr02": Auto-Boost Voltage. • "Pr03": MAX current Limit. • "Pr04": Min current limit. To move between digits, press the down button. To change the digit value, press the up button. To Save the parameter value, press the enter button for 1s. To exit from the setup mode, press up and down button at the same moment.

**Toggling between
12v-24v Operating
Mode**

-for toggle between operating modes (12v-24v) press the **UP Button for 5 sec** and the charger will automatically restart and operate with the desired operating mode.

Models:

	IR-320-12	IR-600-48	IR-1500-24
Wattage	290	580	1500
Input voltage	100 ~ 240VAC	100 ~ 240VAC	180~255Vac
Output Current	20A	10A	60A
Output Voltage	Boost charge 14.4V, Float charge 13.6V	Boost charge 57.6V, Float charge 54.4V	Boost charge 28.8v, Float charge 27.2V
FREQUENCY RANGE	47 ~ 63Hz	47 ~ 63Hz	47 ~ 63Hz
Working Temperature	-30~70°C		
Safety Standards	UL62368-1,TUV BS EN/EN62368-1,EAC TP TC 004, .1,BSMI CNS15598-1, AS/NZS 60950.1,CCC GB4943 IS13252(Part1)/ IEC60950-1(except .5V,48V) Dekra EN 61558-1/2-16,IEC 61558-1/2-16		
EMC Immunity	Compliance to EN61000-4-2,3,4,5,6,8,11, EN55035, light industry level,BS EN/ BS EN/ EAC TP TC 020		

Features:

- 3-stage charging curve for charging mode.
- 30~70° wide operating temperature.
- Multiple protections: short circuit/ Over load/ Over voltage/ Over Temperature.
- Suitable for lead-acid (Pb) batteries.

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